

and help make health-care decisions. As we will see later in the chapter, a *genetic profile* is currently taken to mean a set of genetic markers such as SNPs. Ultimately, however, it will likely mean the complete DNA sequence of an individual—after sequencing becomes inexpensive enough. In fact, a 2016 study suggested that for difficult-to-diagnose patients it might be faster and more economical to sequence just the expressed regions (the exons, together called the *exome*) or even all of the genome right away, rather than running through a standard series of diagnostic tests. The ability to sequence a person's genome quickly and inexpensively is advancing rapidly, perhaps faster than development of appropriate treatments for the conditions. Still, the identification of genes involved in these conditions provides targets for therapeutic interventions.

An individual's genomic information can be used to predict the benefits and risks of particular medications, an approach called *pharmacogenetics*. There are over 300 medications for which the Food and Drug Administration recommends genetic testing for patients prior to their use.

Human Gene Therapy and Gene Editing

Gene therapy—the introduction of genes into an afflicted individual for therapeutic purposes—holds great potential for treating the relatively small number of disorders traceable to a single defective gene. The aim of this approach is to insert a normal allele of the defective gene into the somatic cells of the tissue affected by the disorder.

➔ Mastering Biology BBC Video: Cystic Fibrosis and the Promise of Gene Therapy

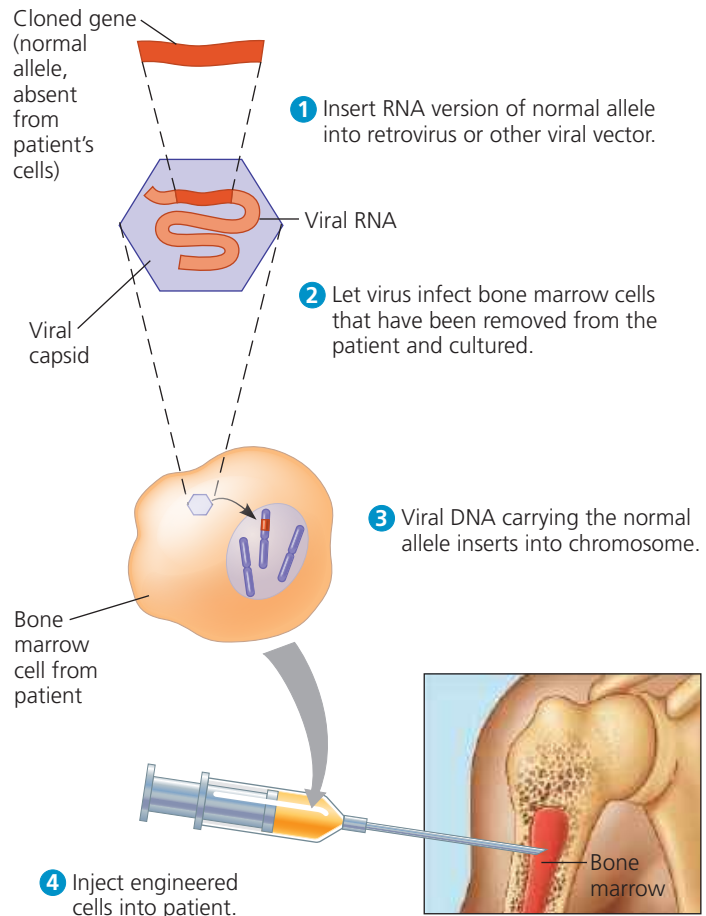
For gene therapy of somatic cells to be permanent, the cells that receive the normal allele must be cells that multiply throughout the patient's life. Bone marrow cells, which include the stem cells that give rise to all the cells of the blood and immune system, are prime candidates.

Figure 20.22 outlines one procedure for gene therapy in an individual whose bone marrow cells do not produce a vital enzyme because of a single defective gene. One type of severe combined immunodeficiency (SCID) is caused by this kind of defect. If the treatment is successful, the patient's bone marrow cells will begin producing the missing protein, and the patient may be cured.

The procedure shown in Figure 20.22 was used in gene therapy trials for SCID in France in 2000, resulting in the first indisputable success of gene therapy. However, three patients subsequently developed leukemia, a type of blood cell cancer, and one of them died. Researchers have concluded it is likely that the insertion of the retroviral vector occurred near a gene that triggers the proliferation of blood cells. Using a viral vector that does not come from a retrovirus, clinical researchers have treated several other genetic diseases somewhat successfully with gene therapy, including

▼ Figure 20.22 Gene therapy using a retroviral vector.

A retrovirus that has been rendered harmless is used as a vector in this procedure, which exploits the ability of a retrovirus to insert a DNA transcript of its RNA genome into the chromosomal DNA of its host cell (see Figure 19.9). If the foreign gene carried by the retroviral vector is expressed, the cell and its descendants will possess the gene product. Cells that reproduce throughout life, such as bone marrow cells, are ideal candidates for gene therapy.



➔ Mastering Biology HHMI Video: Gene Therapy HHMI Video: The Search for a Mutated Gene HHMI Video: Genes as Medicine



a type of progressive blindness (see Concept 50.3), a degenerative disease of the nervous system, and a blood disorder involving the β -globin gene.

Gene therapy raises many technical issues. For example, how can the activity of the transferred gene be controlled so that cells make appropriate amounts of the gene product at the right time and in the right place? How can we be sure that the insertion of the therapeutic gene does not harm some other necessary cell function? As more is learned about DNA control elements and gene interactions, researchers may be able to answer such questions.

A more direct approach that avoids the complications of using a viral vector in gene therapy is made possible by gene editing, especially using the CRISPR-Cas9 system